

Space Shooter - Complete Unity Setup Guide

This guide walks you through setting up the Space Shooter game in Unity Editor step-by-step.

Prerequisites

- **Unity Hub** installed ([Download](https://unity.com/download) (<https://unity.com/download>))
 - **Unity Editor 2021.3 LTS or newer** (2022.3 LTS recommended)
 - **Windows 10/11** for building Windows executables
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Step 1: Open the Project in Unity

1. Open **Unity Hub**
2. Click **“Open”** (or “Add project from disk”)
3. Navigate to and select the `space_shooter_game` folder
4. Unity will detect it as a project and import all assets
5. Wait for the initial import to complete (may take 1-2 minutes)

Note: If Unity asks about the render pipeline, select **Built-in Render Pipeline** (default).

Step 2: Configure Tags and Layers

Unity needs custom tags for collision detection. Go to **Edit → Project Settings → Tags and Layers**:

Tags (add these if not already present):

Tag Name	Purpose
Player	Player ship
PlayerBullet	Bullets fired by the player
EnemyBullet	Bullets fired by enemies
Enemy	All enemy ships
PowerUp	All power-up pickups

Sorting Layers (add in order):

Layer Name	Order
Background	0 (bottom)
Default	1
Foreground	2
UI	3 (top)

Step 3: Configure Sprite Import Settings

For each sprite in `Assets/Sprites/` :

1. Select the sprite file in the Project window
2. In the Inspector, set:
 - **Texture Type:** Sprite (2D and UI)
 - **Pixels Per Unit:** 100
 - **Filter Mode:** Point (for crisp pixel look) or Bilinear
 - **Compression:** None
3. Click **Apply**

Step 4: Create the Main Game Scene

4.1 Create a New Scene

1. Go to **File** → **New Scene** (select "Basic 2D")
2. Save it as `Assets/Scenes/GameScene.unity`

4.2 Set Up the Camera

1. Select **Main Camera** in the Hierarchy
2. Set **Background Color** to dark blue/black: `(8, 8, 20, 255)` or `#08081`
3. Set **Size** to `5.5` (for the orthographic camera)
4. Set **Position** to `(0, 0, -10)`
5. Add the `ScreenBounds` script to the camera

4.3 Create the GameManager Object

1. Create an **Empty GameObject**: `GameObject` → `Create Empty`
2. Name it `GameManager`
3. Add the script: **Add Component** → **SpaceShooter.Managers.GameManager**
4. This is a singleton that persists across scenes

4.4 Create the AudioManager Object

1. Create an **Empty GameObject** named `AudioManager`

2. Add the script: **Add Component** → **SpaceShooter.Managers.AudioManager**
3. Add **two AudioSource** components to it
4. Assign the first AudioSource to the `Sfx Source` field
5. Assign the second AudioSource to the `Music Source` field (check **Loop**)
6. Drag and drop audio clips from `Assets/Audio/` into the corresponding fields:
 - Shoot Clip ← `shoot.wav`
 - Explosion Clip ← `explosion.wav`
 - Power Up Clip ← `powerup.wav`
 - Player Hit Clip ← `player_hit.wav`
 - Wave Start Clip ← `wave_start.wav`
 - Game Over Clip ← `game_over.wav`
 - Button Click Clip ← `button_click.wav`

4.5 Create the SpawnManager Object

1. Create an **Empty GameObject** named `SpawnManager`
2. Add the script: **Add Component** → **SpaceShooter.Managers.SpawnManager**
3. We'll assign enemy prefabs after creating them (Step 5)

Step 5: Create Prefabs

5.1 Player Ship Prefab

1. Create an **Empty GameObject** named `Player`
2. Set **Tag** to `Player`
3. Set **Position** to `(0, -3, 0)`
4. Add **SpriteRenderer**: Assign `player_ship.png` sprite; Sorting Layer: `Default` ; Order: 5
5. Add **BoxCollider2D**: Check `Is Trigger` ; adjust size to fit the ship sprite (~0.5 x 0.5)
6. Add **Rigidbody2D**: Set **Body Type** to `Kinematic` (prevents physics forces)
7. Add script: **SpaceShooter.Player.PlayerController**
8. Create a **child Empty GameObject** named `FirePoint` , position at `(0, 0.4, 0)` (front of ship)
9. Create a **child Empty GameObject** named `ShieldVisual` :
 - Add **SpriteRenderer** with `shield_bubble.png`
 - Disable the `ShieldVisual` object (uncheck the checkbox)
10. In `PlayerController` inspector:
 - Drag `FirePoint` to the `Fire Point` field
 - Drag `ShieldVisual` to the `Shield Visual` field
 - Drag the `SpriteRenderer` to the `Sprite Renderer` field
 - Set `Bullet Prefab` (after creating bullet prefab below)
11. **Drag the Player from Hierarchy into** `Assets/Prefabs/` to create a prefab
12. **Keep one instance in the scene**

5.2 Player Bullet Prefab

1. Create an **Empty GameObject** named `PlayerBullet`
2. Set **Tag** to `PlayerBullet`
3. Add **SpriteRenderer**: Assign `player_bullet.png` ; Sorting Layer: `Default` ; Order: 3
4. Add **BoxCollider2D**: Check `Is Trigger` ; adjust size

5. Add **Rigidbody2D**: Body Type = `Kinematic`
6. Add script: **SpaceShooter.Weapons.BulletController**
 - Speed: `12`
 - Damage: `10`
 - Direction: `(0, 1)` (upward)
7. **Drag into** `Assets/Prefabs/` to create prefab
8. **Delete from scene** (it's spawned at runtime)

5.3 Enemy Bullet Prefab

1. Create an **Empty GameObject** named `EnemyBullet`
2. Set **Tag** to `EnemyBullet`
3. Add **SpriteRenderer**: Assign `enemy_bullet.png` ; Sorting Layer: `Default` ; Order: `3`
4. Add **BoxCollider2D**: Check `Is Trigger`
5. Add **Rigidbody2D**: Body Type = `Kinematic`
6. Add script: **SpaceShooter.Weapons.BulletController**
 - Speed: `6`
 - Damage: `10`
 - Direction: `(0, -1)` (downward)
7. **Drag into** `Assets/Prefabs/` , then **delete from scene**

5.4 Enemy Prefabs (3 types)

For each enemy type, follow these steps:

Basic Enemy

1. Create Empty GameObject named `EnemyBasic`
2. Tag: `Enemy`
3. Add **SpriteRenderer**: `enemy_basic.png` ; Sorting Layer: `Default` ; Order: `4`
4. Add **BoxCollider2D**: `Is Trigger` = `true`
5. Add **Rigidbody2D**: Body Type = `Kinematic`
6. Add script: **SpaceShooter.Enemy.EnemyController**
 - Enemy Type: `Basic`
 - Max Health: `30`
 - Move Speed: `3`
 - Score Value: `100`
 - Fire Rate: `1.5`
7. Create child `FirePoint` at `(0, -0.3, 0)` (front of enemy, facing down)
8. Assign `EnemyBullet` prefab to `Bullet Prefab` field
9. Assign `FirePoint` to the `Fire Point` field
10. Assign power-up prefabs to `Power Up Prefabs` array (after creating them)
11. **Drag into** `Assets/Prefabs/` , delete from scene

Zigzag Enemy

- Same as Basic but use `enemy_zigzag.png`
- Name: `EnemyZigzag`
- Enemy Type: `Zigzag`
- Max Health: `20` , Move Speed: `4` , Score Value: `150` , Fire Rate: `1.0`

Tank Enemy

- Same as Basic but use `enemy_tank.png`
- Name: `EnemyTank`
- Enemy Type: `Tank`
- Max Health: `80` , Move Speed: `1.5` , Score Value: `300` , Fire Rate: `2.5`

5.5 Power-Up Prefabs (3 types)

For each power-up:

Health Pack

1. Create Empty GameObject named `PowerUpHealth`
2. Tag: `PowerUp`
3. Add **SpriteRenderer**: `powerup_health.png` ; Order: 6
4. Add **CircleCollider2D**: Is Trigger = true
5. Add **Rigidbody2D**: Body Type = `Kinematic`
6. Add script: **SpaceShooter.PowerUps.PowerUpController**
 - Power Up Type: `HealthPack`
 - Heal Amount: `30`
7. **Drag into** `Assets/Prefabs/` , delete from scene

Rapid Fire

- Same setup, use `powerup_rapidfire.png`
- Name: `PowerUpRapidFire`
- Power Up Type: `RapidFire`

Shield

- Same setup, use `powerup_shield.png`
- Name: `PowerUpShield`
- Power Up Type: `Shield`

5.6 Assign Cross-References

Now go back and connect the prefabs:

1. **Player prefab** → Set `Bullet Prefab` to `PlayerBullet` prefab
2. **All Enemy prefabs** → Set `Bullet Prefab` to `EnemyBullet` prefab
3. **All Enemy prefabs** → Set `Power Up Prefabs` array to include all 3 power-up prefabs
4. **SpawnManager** → Set:
 - `Basic Enemy Prefab` → `EnemyBasic`
 - `Zigzag Enemy Prefab` → `EnemyZigzag`
 - `Tank Enemy Prefab` → `EnemyTank`

Step 6: Create the UI (Canvas)

6.1 Create Canvas

1. **GameObject** → **UI** → **Canvas**
2. Set Canvas Scaler:
 - **UI Scale Mode**: Scale With Screen Size

- **Reference Resolution:** 1920 x 1080
- **Match:** 0.5

3. Name it `UICanvas`

6.2 Create UIManager

1. Create an **Empty GameObject** named `UIManager` (can be child of Canvas or separate)
2. Add script: **SpaceShooter.UI.UIManager**

6.3 Main Menu Panel

1. Under Canvas, create **UI → Panel** named `MainMenuPanel`
2. Set background to semi-transparent dark: `(10, 10, 30, 230)`
3. Add children:
 - **UI → Text** named `TitleText` : "SPACE SHOOTER", font size 72, centered, white, bold
 - **UI → Text** named `HighScoreMenuText` : "High Score: 0", font size 24
 - **UI → Button** named `StartButton` : Text = "START GAME", font size 32
 - **UI → Button** named `QuitButton` : Text = "QUIT", font size 24

6.4 HUD Panel

1. Create **UI → Panel** named `HUDPanel` (set alpha to 0 for transparent background)
2. Add children:
 - **UI → Text** named `ScoreText` : "Score: 0", top-left anchor, font size 28, white
 - **UI → Text** named `WaveText` : "Wave 1", top-center, font size 24
 - **UI → Slider** named `HealthBar` : bottom-left, set colors (fill = green), width ~300
 - **UI → Text** named `HealthText` : "100 / 100", near health bar
 - **UI → Image** named `ShieldIcon` : small blue icon (use shield sprite), top-right area
 - **UI → Image** named `RapidFireIcon` : small orange icon, next to shield icon
3. Start with HUD Panel **disabled** (unchecked)

6.5 Game Over Panel

1. Create **UI → Panel** named `GameOverPanel` (semi-transparent dark background)
2. Add children:
 - **UI → Text** named `GameOverTitleText` : "GAME OVER", font size 64, red, centered
 - **UI → Text** named `FinalScoreText` : "Score: 0", font size 36
 - **UI → Text** named `FinalWaveText` : "Wave Reached: 1", font size 28
 - **UI → Text** named `HighScoreText` : "High Score: 0", font size 24
 - **UI → Text** named `NewHighScoreText` : "NEW HIGH SCORE!", font size 32, yellow (start disabled)
 - **UI → Button** named `RestartButton` : Text = "PLAY AGAIN", font size 28
 - **UI → Button** named `MenuButton` : Text = "MAIN MENU", font size 24
3. Start with GameOver Panel **disabled**

6.6 Pause Panel

1. Create **UI → Panel** named `PausePanel` (semi-transparent)
2. Add children:
 - **UI → Text**: "PAUSED", font size 48, centered
 - **UI → Button** named `ResumeButton` : Text = "RESUME"
 - **UI → Button** named `PauseMenuButton` : Text = "MAIN MENU"
3. Start **disabled**

6.7 Wave Announcement Panel

1. Create **UI → Panel** named `WaveAnnouncementPanel` (transparent bg)
2. Add **UI → Text** named `WaveAnnouncementText` : “WAVE 1”, font size 56, centered, white with outline
3. Start **disabled**

6.8 Wire Up UIManager

Select the `UIManager` object and drag all the panels and UI elements into the corresponding fields in the Inspector:

- Main Menu Panel, HUD Panel, Game Over Panel, Pause Panel, Wave Announcement Panel
- All Text fields, Buttons, Slider, Images as labeled

Step 7: Set Up Background

7.1 Star Field (Procedural)

1. Create an **Empty GameObject** named `StarField`
2. Add script: **SpaceShooter.Environment.StarField**
3. Set Position to `(0, 0, 5)` (behind everything)
4. Configure: Star Count = 200, Field Width = 20, Field Height = 15

7.2 Parallax Background (Optional - for richer visuals)

1. Create an **Empty GameObject** named `ParallaxBackground`
2. Add script: **SpaceShooter.Environment.ParallaxBackground**
3. Create two child GameObjects:
 - `BG_Layer1` : Add SpriteRenderer with `bg_layer1.png` , Sorting Layer: `Background` , Order: 0, Scale to fill screen
 - `BG_Layer2` : Add SpriteRenderer with `bg_layer2.png` , Sorting Layer: `Background` , Order: 1, Scale to fill screen
4. Assign the SpriteRenderers to the script's Layer1/Layer2 fields

Step 8: Final Scene Hierarchy Check

Your Hierarchy should look like:

```

GameScene
├── Main Camera (+ ScreenBounds script)
├── GameManager (+ GameManager script)
├── AudioManager (+ AudioManager script + 2x AudioSource)
├── SpawnManager (+ SpawnManager script)
├── Player (+ PlayerController, SpriteRenderer, BoxCollider2D, Rigidbody2D)
│   ├── FirePoint
│   └── ShieldVisual (disabled)
├── StarField (+ StarField script)
├── ParallaxBackground (+ ParallaxBackground script)
│   ├── BG_Layer1
│   └── BG_Layer2
├── UICanvas (Canvas + Canvas Scaler + Graphic Raycaster)
│   ├── UIManager (+ UIManager script)
│   ├── MainMenuPanel
│   │   ├── TitleText
│   │   ├── HighScoreMenuText
│   │   ├── StartButton
│   │   └── QuitButton
│   ├── HUDPanel (disabled)
│   │   ├── ScoreText
│   │   ├── WaveText
│   │   ├── HealthBar (Slider)
│   │   ├── HealthText
│   │   ├── ShieldIcon
│   │   └── RapidFireIcon
│   ├── GameOverPanel (disabled)
│   │   ├── GameOverTitleText
│   │   ├── FinalScoreText
│   │   ├── FinalWaveText
│   │   ├── HighScoreText
│   │   ├── NewHighScoreText
│   │   ├── RestartButton
│   │   └── MenuButton
│   ├── PausePanel (disabled)
│   │   ├── PausedText
│   │   ├── ResumeButton
│   │   └── PauseMenuButton
│   └── WaveAnnouncementPanel (disabled)
│       └── WaveAnnouncementText
└── EventSystem (auto-created with Canvas)

```

Step 9: Test in Editor

1. Press **Play** in the Unity Editor
2. You should see:
 - Main menu with “START GAME” button
 - Click Start → HUD appears, player ship at bottom
 - Arrow keys / WASD to move, Space to shoot
 - Enemies spawn from top
 - Killing enemies gives score
 - Power-ups drop occasionally
 - Taking damage reduces health bar
 - Health = 0 → Game Over screen
 - ESC key pauses/unpauses

Step 10: Build for Windows (.exe)

1. Go to **File → Build Settings**
2. Ensure your `GameScene` is in the **Scenes In Build** list
 - If not, click **Add Open Scenes**
3. Select **Platform: PC, Mac & Linux Standalone**
4. Set **Target Platform:** Windows
5. Set **Architecture:** x86_64
6. Click **Player Settings** and configure:
 - **Product Name:** Space Shooter
 - **Company Name:** (your name)
 - **Default Screen Width:** 1024
 - **Default Screen Height:** 768
 - **Fullscreen Mode:** Windowed (or Fullscreen Window)
7. Click **Build** (or **Build And Run**)
8. Choose an output folder (e.g., `Builds/Windows`)
9. Wait for the build to complete
10. Your `.exe` will be in the output folder along with a `_Data` folder

Distributing the Build

To share your game, zip the entire output folder containing:

- `Space Shooter.exe`
- `Space Shooter_Data/` folder
- `UnityPlayer.dll`
- `MonoBleedingEdge/` folder

All files must stay together for the game to work.

Troubleshooting

Issue	Solution
Scripts won't compile	Ensure all script files are in <code>Assets/Scripts/</code> subfolders
Tags not found error	Manually add tags in Edit → Project Settings → Tags and Layers
Collisions not working	Ensure all colliders have <code>Is Trigger</code> checked and <code>Rigidbody2D</code> exists
Bullets not spawning	Check that bullet prefab is assigned in <code>PlayerController</code> and <code>EnemyController</code>
UI not showing	Ensure Canvas has <code>EventSystem</code> child, and <code>UIManager</code> fields are all assigned
No sound	Ensure <code>AudioManager</code> has <code>AudioSource</code> components and clips are assigned
Enemies not spawning	Check <code>SpawnManager</code> has all 3 enemy prefabs assigned
Power-ups not appearing	Check enemy prefabs have power-up prefabs in their array

Controls Summary

Key	Action
Arrow Keys / WASD	Move ship
Spacebar	Fire weapon
ESC	Pause / Unpause